

N+- Network Fundamentals and building network

[Q] Introduction to IP Addressing Theory (1 hour):IPv4 and IPv6: Differences, structure, and importance. Subnetting basics: CIDR, subnet masks. Practical (1 hour): Assign static IPs to devices and verify connectivity. Task: Solve subnetting exercises and calculate valid subnets.

[Q-1] List the key differences between IPv4 and IPv6 addressing by creating a two-column table with at least 4 points for each.

[ANS]

Feature	IPv4	IPv6
Address Length	32-bit	128-bit
Address Format	Decimal (e.g., 192.168.1.1)	Hexadecimal (e.g., 2001:db8::1)
Number of Addresses	About 4.3 billion	About 340 undecillion
Configuration	Manual or DHCP	Auto-configuration supported
Security	Optional IPsec	Built-in IPsec support
Broadcast Support	Supports Broadcast	No Broadcast, uses Multicast

[Q-2] Given the following scenario: You have 4 devices (laptop, phone, smart TV, and printer) connected to your home WiFi. Assign a unique static IPv4 address to each device using the 192.168.10.0/24 network, and write down the IP, subnet mask, and default gateway for each device.

[ANS]

- Network: 192.168.10.0/24
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1

Device	IP Address	Subnet Mask	Default Gateway
Laptop	192.168.10.10	255.255.255.0	192.168.10.1
Phone	192.168.10.20	255.255.255.0	192.168.10.1
Smart TV	192.168.10.30	255.255.255.0	192.168.10.1
Printer	192.168.10.40	255.255.255.0	192.168.10.1

[Q-3] Calculate the number of valid subnets and hosts per subnet if you use a /27 CIDR subnet mask on a 192.168.1.0 network. List the first three valid subnet address ranges (network address to broadcast address) and the usable host IPs in each.

[ANS]

○ **CIDR Information**

- Network Address: 192.168.1.0
- CIDR: /27
- Subnet Mask: 255.255.255.224

○ **Calculations**

- Number of Host Bits = $32 - 27 = 5$
- Hosts per Subnet = $2^5 - 2 = 30$ usable hosts

○ **First Three Valid Subnets**

Subnet	Network Address	First Host	Last Host	Broadcast Address
Subnet 1	192.168.1.0	192.168.1.1	192.168.1.30	192.168.1.31
Subnet 2	192.168.1.32	192.168.1.33	192.168.1.62	192.168.1.63
Subnet 3	192.168.1.64	192.168.1.65	192.168.1.94	192.168.1.95

[Q-4] You are given the IP address 10.0.5.17 with subnet mask 255.255.255.0. Determine the network address, broadcast address, and the range of valid host addresses for this subnet. Hint: Convert the subnet mask to CIDR notation and use bitwise AND for network address calculation.

[ANS]

○ Step 1: Convert Subnet Mask to CIDR

- Subnet Mask = 255.255.255.0
- CIDR Notation = /24

○ Step 2: Determine Network Information

Item	Value
IP Address	10.0.5.17
Subnet Mask	255.255.255.0
CIDR	/24
Network Address	10.0.5.0
Broadcast Address	10.0.5.255
First Valid Host	10.0.5.1

Last Valid Host	10.0.5.254
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○ **Result**

- Network Address: 10.0.5.0
- Broadcast Address: 10.0.5.255
- Valid Host Range: 10.0.5.1 – 10.0.5.254

[Q-5] Use an online subnet calculator tool (such as <https://www.subnet-calculator.com/>) to verify your answer for the previous task. Take a screenshot of your calculation and write one line explaining if your manual answer matched the tool's output.

[ANS]

○ **Subnet Calculator Used:** <https://www.subnet-calculator.com/>

○ **Verification Result:**

The subnet calculator confirmed that the network address is **10.0.5.0**, the broadcast address is **10.0.5.255**, and the valid host range is **10.0.5.1 to 10.0.5.254**.

IPv4 Subnet Calculator

Result

IP Address:	192.168.0.110
Network Address:	192.168.0.108
Usable Host IP Range:	192.168.0.109 - 192.168.0.110
Broadcast Address:	192.168.0.111
Total Number of Hosts:	4
Number of Usable Hosts:	2
Subnet Mask:	255.255.255.252
Wildcard Mask:	0.0.0.3
Binary Subnet Mask:	11111111.11111111.11111111.11111100
IP Class:	C
CIDR Notation:	/30
IP Type:	Private
Short:	192.168.0.110 /30
Binary ID:	110000001010100000000000001101110
Integer ID:	3232235630
Hex ID:	0xc0a8006e
in-addr.arpa:	110.0.168.192.in-addr.arpa
IPv4 Mapped Address:	::ffff:c0a8.06e
6to4 Prefix:	2002:c0a8.06e::/48